Assignment 6: Implementation of Data Integrity Constraints

Keys, validation rules and verification

Project Topic: Hotel Management System

1. Objective

Data integrity constraints keep records valid and maintain correct relationships between tables. The schema in Assignment 5 includes primary keys, foreign keys, UNIQUE, CHECK, NOT NULL and DEFAULT constraints.

2. Constraints Used

Constraint	Where used	Purpose
PRIMARY KEY	Every table's ID column	Uniquely identifies each row and cannot be NULL
FOREIGN KEY	Rooms, Reservations, Payments, ServiceOrders	Prevents references to missing parent records
UNIQUE	Guests.email, Guests.id_proof_number, Rooms.room_number, RoomTypes.type_name, Services.service_name	Prevents duplicate values where uniqueness is required
NOT NULL	Required names, IDs, dates, amounts and relationship columns	Requires essential data to be supplied
CHECK	Prices, occupancy, quantities, dates and status values	Rejects invalid values such as negative prices or invalid status labels
DEFAULT	Room status, staff shift, booking status, payment status, service quantity	Provides a standard value when one is not supplied

3. Verification Queries

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-- Valid insert: room type exists and room number is unique\nINSERT INTO Rooms (room_id, room_number, floor_number, room_typ
```

4. Expected Verification Results

Test	Expected outcome	Reason
Insert room with valid room type	Succeeds; status defaults to Available	DEFAULT applies when status is omitted
Duplicate room number	Rejected	UNIQUE constraint
Room references room_type_id 999	Rejected	Foreign key requires an existing RoomTypes row
Negative room price	Rejected	CHECK requires price >= 0
Checkout date before check-in	Rejected	CHECK requires checkout later than check-in

Note: CHECK constraint enforcement requires a modern MySQL version (MySQL 8.0.16 or later). Run each invalid test separately because a failed statement stops that individual statement, not the whole database.

5. Conclusion

Primary and foreign keys protect identity and relationships, while UNIQUE, CHECK, NOT NULL and DEFAULT constraints prevent common data errors. The verification statements demonstrate both accepted and rejected data.